Simulating Random Walks on the Hyperbolic Plane

Alex Shi

Swarthmore College

Objectives

In this project, we simulated random walks on the hyperbolic plane to examine its properties. We focused on quasi-geodesic paths that don't double back on themselves. We are interested in quantitative estimates of the quasi-geodesic constants of such paths, how these constants may vary, and by how much. We want to get a good quantitative read on what effect the constants have by examining the statistics of the random walks.

Introduction

Since hyperbolic space expands exponentially, we expect that small local differences between some paths in hyperbolic space should lead to those paths having necessarily different endpoints. Thus while two walkers on the Euclidean plane can always meet with each other without doubling back on their path, there are many paths where two walkers on the hyperbolic plane can no longer reach each other without doubling back. Therefore at extreme length, two paths with the same endpoint must remain within some distance D of each other that does not depend on the choice of paths.

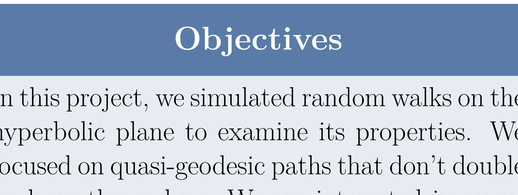


Figure 1: Paths on the Hyperbolic plane may not be able to reach each other.

We simulated random walks with a straight path interspersed with "upward" jumps, i.e. a sort of stair pattern. Using a simulation based on the Poincaré half-plane model written in MATLAB, we varied the total length of the path, the number of jumps, and the minimum length of each step of the random walks to create quasi-geodesic paths in $\mathbb{H}^2$.

Definitions

We use the Poincaré half-plane model to model two-dimensional hyperbolic geometry in the upper half-plane. In this model, every geodesic (shortest path between two points) is either a subsegment of a vertical line or a semicircle whose endpoints lie on the x-axis. The length of a path $\gamma(t) = (x(t), y(t))$ can be calculated by:

$$\text{dist}(a, b) = \int_{t_1}^{t_2} \sqrt{\frac{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2}{y^2}} dt$$

We can find the distance between a and b by finding integrating over a geodesic between them.

Methodology

The random walks we studied consisted of a straight (geodesic) walk at a constant speed with "jumps" that move the walker to the right of its position. This was simulated as geodesic segments that alternate turning right and left.

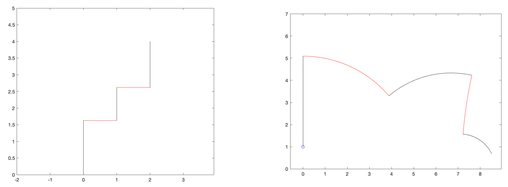


Figure 2: Similar walks on Euclidean and Hyperbolic planes.

There were four main variables we varied to analyse the random walks: the number of jumps the walker takes, the minimum amount of distance before the walker can jump again, the total length of the walk, and the maximum distance the walker can jump. Due to the nature of the Poincaré half-plane model, most of the space is "squished" near the x-axis. Thus, long paths tend to be projected very close to the x-axis where we lose floating point precision as the paths get more compressed. Thus we focused on walks with fewer than 50 jumps, where the simulation is still precise.

Results

For each simulation we plotted and examined the ratio between the geodesic and "walking" distance, and the geodesic distance itself.

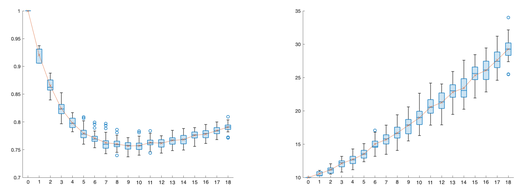


Figure 3: The ratio and the distance from 0 to 18 jumps.

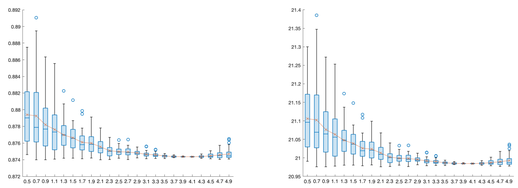


Figure 4: The ratio and the distance from a minimum step distance of 0.5 to 4.9.

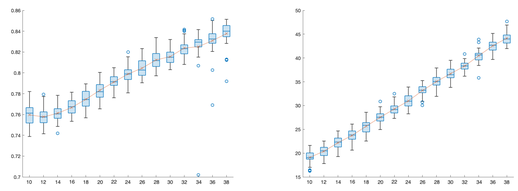


Figure 5: The ratio and the distance from a total walk length of 10 to 38.

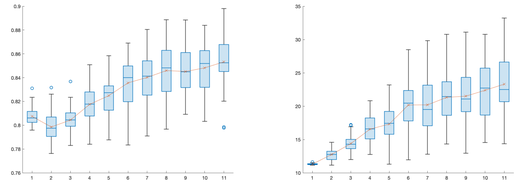


Figure 6: The ratio and the distance from a maximum jump length of 1 to 11.

Conclusion

We observe that the ratio appears to be approximately between 80% and 90% no matter the conditions. This suggests the endpoint of the walks can be used as a decent (although not exact) proxy to calculate the number of jumps. From figures 3 and 4, we hypothesize that the variation in the first step leads to a large variance initially, but as the number of jumps increases, the variation gets averaged out. Also, each jump appears to add a constant amount of distortion from being geodesic while long straight segments make the path closer to being geodesic. From figures 5 and 6, we can see that as the individual parts of the paths get longer (walk length or jump length), the distance between the start and end points of the path also get longer.

Additional Questions

Some further questions that could be investigated in the future:

- How much does the endpoint tell us about the original walk?
- For dimensional reasons, it's not possible to fully recover a unique walk from an endpoint, but are there any restrictions on the family of walks for a particular endpoint?
- Are there interesting things that the jump times alone might tell us about the trajectory of the walk?

Acknowledgements

This project was funded by the Surdna Summer Research Fellowship. Thank you to Professor Eduard Einstein for guiding and advising me this summer.

SWARTHMORE